Secure and Role-Aware Explainable Occupant Tracking in Multi-Building Indoor Smart Spaces

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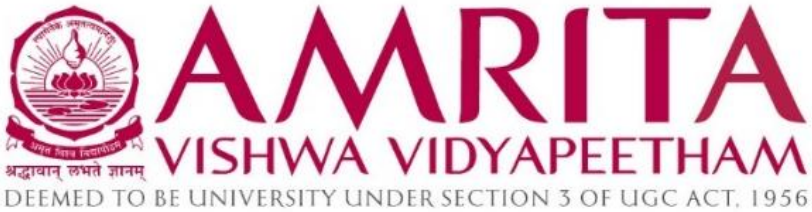
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BONAFIDE CERTIFICATE

This is to certify that the project report entitled "Secure and Role-Aware Explainable Occupant Tracking in Multi-Building Indoor Smart Spaces", submitted by Ajith Ashok (AM.U4.U4CSE23204), Arjun Rajesh (AM.U4.U4CSE23208), D Aditya Kiran (AM.U4.U4CSE23215) and Yadhu Vipin Madhilakath Madathil (AM.U4.U4CSE23270), in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering at Amrita Vishwa Vidyapeetham, is a bonafide record of the work carried out by them under my guidance and supervision at Amrita School of Computing, Amritapuri, during Semester 7 of the academic year 2026 - 2027.

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DECLARATION

We, Ajith Ashok (AM.U4.U4CSE23204), Arjun Rajesh (AM.U4.U4CSE23208), D Aditya Kiran (AM.U4.U4CSE23215) and Yadhu Vipin Madhilakath Madathil (AM.U4.U4CSE23270), hereby declare that the project entitled "Secure and Role-Aware Explainable Occupant Tracking in Multi-Building Indoor Smart Spaces" is a record of our original work carried out under the guidance of Dr. Lakshmi Mohan [Guide designation], Department of Computer Science and Engineering, Amrita Vishwa Vidyapeetham, Amritapuri. To the best of our knowledge, this work has not formed the basis for the award of any degree, diploma, fellowship or similar qualification to any candidate at any university. All sources and contributions of others have been appropriately acknowledged.

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ABSTRACT

Multi-building campuses need to recognise and follow people across buildings, but keeping every face template in one central store concentrates sensitive biometric data, while keeping the templates apart makes it hard to find the building that knows a visitor. This project extends the Distributed State Transition System (DSTS) framework to decentralised multi-building occupant identification and tracking, with secure, role-based and explainable access to the results. Each building keeps its own gallery and publishes only a Bloom filter built from locality-sensitive-hash (LSH) codes of its occupants' face embeddings. A capture that is not recognised locally is routed to a short list of likely buildings, which verify it by angular-distance voting. A dynamic visitor database, per-state probabilistic tracking, two-layer role-based access control with a class-teacher model, sealed inter-building transfers, and explanations redacted to the caller's disclosure level complete the system. It is evaluated on 550 simulated occupants in 11 buildings (500 from CASIA-WebFace and 50 masked people from the real-world RMFRD dataset) over 11,000 events per simulated day, against a centralised search. On seed 42 the decentralised pipeline identified 91.76% of events against 92.24% for the central search while touching 10.3% of the galleries; over five random populations accuracy was 91.54% ± 0.20 (95% CI) and building-level state F1 was 0.9501. Unmasked people were identified 95.0% of the time and masked people 59.0%; synthetic-masked templates, upper-face matching and relaxed thresholds did not improve masked F1. 571 automated tests cover the policy, secure-transfer and explanation layers. The network, cameras and certificate authority are simulated, so the work demonstrates the design and its accuracy rather than a hardened deployment.

Keywords: Occupant identification and tracking; Distributed state transition system; Locality-sensitive hashing; Bloom filter; Role-based access control; Explainable queries; Masked face recognition

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Chapter 1
Introduction

1.1 Background and motivation

Smart spaces use cameras, sensors and computing to observe who is where and to support occupant-specific services such as access, scheduling and safety [6], [7], [26]. On a university campus or a corporate estate these services span many buildings, so the system must keep identifying and following a person as they walk from one building to the next. Camera networks for indoor tracking are well studied [1]-[5], but the identity question changes when the space has several buildings: a person registered in one building is a *visitor* in every other, and the building that captures the face does not hold that person's reference photos.

There are two obvious designs. A *centralised* design keeps every person's reference embeddings in one store and searches all of them for every capture. It is simple and accurate, but it concentrates biometric data, which is the data that cannot be reissued if it leaks, and it makes one server the single point of failure and of attack. A *distributed* design keeps each building's reference data in that building. This respects the data's locality, but a visitor's home building is unknown, so the capturing building must either contact every other building for every unrecognised face or share reference embeddings between buildings [6]. The base paper that this project extends, the Distributed State Transition System (DSTS) of Mohan and Menon [6], resolves visitors through a central master list.

A second gap concerns the people who ask questions of the tracking data. The same data that lets a Dean find a lost visitor also lets any user follow any other person, so the system must decide who may see whom, and at what precision, and must be able to show why an answer was given. Access control for tracking queries and explanations of the answers are rarely treated together with distributed identification.

This project therefore combines three things in one system: (i) decentralised identification and tracking in which only a one-way Bloom filter leaves a building until a match is confirmed; (ii) role-based, privacy-aware querying with a two-layer policy; and (iii) explanations that justify both the access decision and the tracking evidence behind each answer.

Relevant Sustainable Development Goals. The work maps to three goals of the 2030 Agenda [27]. *SDG 9 (Industry, Innovation and Infrastructure)*, target 9.1 on quality, reliable and resilient infrastructure: the system is a resilient, decentralised infrastructure for smart buildings with no central biometric store to lose. *SDG 11 (Sustainable Cities and Communities)*, target 11.7 on safe and inclusive spaces: occupants of a campus can be located for safety and service while the data is protected from misuse such as stalking. *SDG 16 (Peace, Justice and Strong Institutions)*, target 16.6 on effective, accountable and transparent institutions: every access decision is audited and every answer is explained. The intended beneficiaries are campus administrators, who get reliable tracking, and the occupants, whose biometric data stays in their own building and whose location is visible only to people with a legitimate relationship to them.

1.2 Objectives

The objectives, each linked to how it is evaluated and to its deliverable, are given in Table 1.

Table 1: Objectives, evaluation methods and deliverables

#	Objective	Evaluation method	Deliverable
1	Extend the DSTS framework to distributed multi-building identification and tracking	Event identification precision / recall; state-level precision / recall over every state	Pipeline, building nodes, BSTS state store
2	Provide an RBAC query interface for Dean, Teacher and Student roles	Every policy cell tested; access matrix checked against expected decisions	Two-layer policy, FastAPI server, dashboard
3	Return explainable responses that justify the access decision and the tracking evidence	Faithfulness and no-leakage tests on every query class	Explanation model, redaction, “Why this answer” card
4	Design LSH + Bloom-filter routing for decentralised visitor lookup	Routing recall; gallery touched per lookup; accuracy against a centralised search	Enrolment, routing and verification modules
5	Secure the system: sealed transfers, role-based filtering, audit logging	Tamper, replay, wrong-sender and fail-closed tests; login lockout tests	secure_transfer and access_control packages
6	Evaluate tracking precision-recall, storage / communication cost and explanation quality	TP / FP / FN / TN per cohort and level; five seeds; centralised baseline	Evaluation scripts and result tables

1.3 Scope and constraints

Intended users. Campus administrators (Deans), faculty (Teachers) and students; each sees tracking data only as their role and relationship allow. **Included functions.** Face-embedding based identification of registered occupants and of visitors across buildings; probabilistic zone-level tracking; thirteen query templates (Q1-Q13) and a full-trajectory

query; role-based disclosure; explanations; secure inter-building exchange; a web dashboard; and a centralised baseline for comparison.

Exclusions and assumptions. The system works on face embeddings that are produced by a pretrained model; it does not train a face model and does not process live video. Cameras, the network between buildings and the certificate authority are simulated in one process. The occupant population is synthetic in its movements: 20 reference and 20 test photos per person come from public datasets, and the test photos are replayed as camera events. Zones are a fixed nine-zone layout per building. **Constraints.** All experiments run on a laptop CPU; every result is reproducible from a population seed.

1.4 Project report organization

Chapter 2 states the problem and the limits of existing systems. Chapter 3 reviews fourteen related works and compares them. Chapter 4 lists the functional, non-functional, hardware and software requirements. Chapter 5 presents the design and the step-by-step approach, including the algorithms and parameters. Chapter 6 describes the evaluation, reports the results with tables and figures, compares them with a centralised baseline, and discusses limitations. Chapter 7 concludes and lists future work. The references follow.

Chapter 2
Problem Definition

2.1 Problem statement

Problem. Identify and track occupants across a wide-area, multi-building indoor environment while giving secure, role-based and explainable access to where they are and have been, and justify both the access decisions and the tracking evidence behind each answer.

Stakeholders and consequences. Occupants want their biometric data and their movements protected; administrators want reliable tracking and the ability to find a visitor; faculty need to follow their own students; auditors need to know who asked what. If identification is centralised, one breach exposes everyone's biometric data. If it is distributed without routing, every unrecognised face costs a request to every building. If access is not controlled, tracking becomes a stalking tool. If answers are not explained, they cannot be trusted or audited.

Inputs and outputs. The identification input is a face embedding captured at a building, zone and time; the output is an identity (or an abstention) and an updated probability of the person being in each zone. The query input is a request from a logged-in user (role, target, time or interval, purpose); the output is the permitted answer together with its explanation, or a refusal.

2.2 Existing system and limitations

Existing indoor tracking systems use cameras, radio signals or both [1], [2], [9]. Edge-camera frameworks localise and track people accurately inside one site [1], [2], [5], and distributed trackers associate targets across cameras with consensus filtering or shared target tables [3], [4]. These systems stay within one site. They do not address how the identity of a person who is registered elsewhere is discovered, and they do not define who may query the resulting locations.

The DSTS framework [6] models a multi-building environment as states, events and transition functions and resolves visitors through a centralised master list. Bloom filters have been used to protect biometric templates [10], and a recent privacy-preserving crowd-counting system combines face features, fuzzy extractors and encrypted Bloom filters [11]. Summary Cache [13] uses Bloom filters to find the web proxy likely to hold an object.

None of them combines approximate, similarity-based routing of a face to a building with role-based access control and explanations.

Gap. Four gaps remain: (i) identity discovery across separate buildings without sharing galleries; (ii) evidence that LSH with Bloom filters can route biometric visitors accurately; (iii) role-based control over tracking queries; and (iv) explanations for access decisions and tracking answers. This project addresses all four.

Chapter 3
Literature Review

3.1 Review scope and selection

Fourteen works ([1]-[6], [8], [10]-[16]) were reviewed in eleven groups (Section 3.2). The search covered indoor occupant tracking with camera and RF sensing, multi-camera association, state-transition models of smart spaces, privacy-preserving biometrics and Bloom-filter indexing, and similarity search with locality-sensitive hashing. Search keywords were: “indoor localization camera network edge”, “distributed multi-camera multi-target tracking”, “smart space state transition”, “Bloom filter biometric”, “locality sensitive hashing” and “masked face recognition dataset”. The publication period is 1970 to 2026: recent works (2020-2026) on tracking and privacy-preserving biometrics were chosen for the problem context, and foundational works on Bloom filters and hashing were included because the routing mechanism is built on them. A work was selected if it is a primary source (a journal article, conference paper or preprint), is directly relevant to tracking, identity discovery, hashing-based routing, or the models used in this project, and gives enough detail to describe its approach and evaluation. Papers [1]-[5] are the mid-review set; [6] is the base paper that this work extends.

3.2 Descriptive review of literature works

3.2.1 Group identification and localization with privacy-preserving edge cameras

Hegde et al. (2024) in “Indoor group identification and localization using privacy-preserving edge computing distributed camera network” [1] investigate how to find when and where groups form in an indoor therapeutic space of about 1,700 m². The work proposes a pipeline that takes noisy positions and orientations of people, estimated from sparsely placed multi-view edge cameras running a lightweight two-dimensional pose detector, and groups them. The study is situated in a therapeutic built environment with seven participants over 35-minute sessions in groups of two to seven. The authors assess it with precision, recall and F1 for group identification and mean absolute error for group localization. They report F1 of up to about 90% and a localization error of 1.25 m, with raw images kept off the network because the edge devices extract the features. The limitations are that the unit of identification is a group and not an individual, and that the

study covers one space; it says nothing about identifying a person who is registered elsewhere.

3.2.2 Indoor localization and multi-person tracking with sparse edge cameras

Kwon et al. (2023) in "A feasibility study on indoor localization and multiperson tracking using sparsely distributed camera network with edge computing" [2] ask whether people can be localized and tracked over a large indoor site with few, low-cost cameras. They deploy 39 edge camera units, each running pose estimation on a Google Coral accelerator, in a therapeutic space of roughly 1,700 m², and track people from the pose output. The evaluation reports localization error, multiple-object tracking accuracy, tracking precision, identity switches and body-orientation error, with an average localization error of about 1.4 m and a tracking accuracy of about 88.6%. The framework tracks anonymous people inside one site and does not attempt cross-building identity discovery, access control or explanations.

3.2.3 Distributed multi-camera multi-target association

Yang et al. (2022) in "Distributed multi-camera multi-target association for real-time tracking" [3] address real-time association of targets across cameras without a central server. They propose a target-management module in which cameras share a data structure of target labels and appearance information so that detections in different views can be matched, without camera calibration. The evaluation uses several multi-camera benchmarks and an own mobile-camera dataset and reports multi-camera tracking accuracy, IDF1, false positives, false negatives and identity switches. The work improves association, but it keeps no probabilistic state of who is in which building and does not identify where a visitor comes from.

3.2.4 Distributed multi-target tracking in camera networks

Casao et al. (2021) in "Distributed multi-target tracking in camera networks" [4] study tracking of several people with a network of cameras that exchange information only with neighbours. Their approach combines a distributed-consensus Kalman filter with a re-identification network and a distributed tracker manager that handles cross-camera association. They compare against centralised trackers on public multi-camera datasets using tracking accuracy, IDF1-type scores and bandwidth, and report better accuracy and lower bandwidth than the centralised methods they compare with. The gap is that the system tracks within one camera network and has no model of buildings, registered occupants and visitors.

3.2.5 Real-time multi-person multi-camera tracking at the edge

Gaikwad and Karmakar (2021) in “Smart surveillance system for real-time multi-person multi-camera tracking at the edge” [5] build an end-to-end pipeline of detection, tracking and re-identification that runs on an NVIDIA Jetson TX2 edge device. It is evaluated on the authors' own multi-camera campus recordings, collected over a full day, and reports an IDF1 score of 90.97, about 30 frames per second and a re-identification latency of roughly 90 ms. The pipeline shows that edge tracking and re-identification are practical, but it has no formal multi-building state or query model and no decentralised handling of visitors.

3.2.6 Modelling large-scale camera networks: the DSTS framework

Mohan and Menon (2020) in “Modelling large scale camera networks for identification and tracking: an abstract framework” [6] model identification and tracking in a wide-area camera network as a distributed state transition system. A state records the probability that each occupant is in each zone, an event is a recognition at a zone, and a transition function updates the state when an event occurs; each building has a building-level state transition system (BSTS). Registered occupants are recognised against their own building's database, and visitors are resolved through a centralised master list. The framework is abstract and the visitor step is the part this project replaces: the central list concentrates biometric data and every visitor lookup depends on it. The model of states, events and transitions builds on earlier work on the three Rs of cyber-physical spaces [7].

3.2.7 Face recognition and tracking for security surveillance

Nair et al. (2021) in “Face recognition and tracking for security surveillance” [8] investigate identifying and following faces across surveillance cameras, including cameras with non-overlapping views. The work combines face detection and recognition with tracking so that a person recognised in one camera can be followed into the next. It shows that face recognition is a practical identity cue for tracking, which is the assumption this project relies on. As a surveillance system it is not designed around separate per-building galleries, cross-building identity discovery or role-based querying.

3.2.8 Bloom filters for iris biometrics

Rathgeb et al. (2014) in “On application of Bloom filters to iris biometrics” [10] investigate how Bloom filters can hold iris templates. Representing the template as a Bloom filter makes comparison alignment-free and one-way, so that the stored template protects the biometric while still allowing identification, and the structure supports faster searches than a linear scan of templates. The work establishes that Bloom filters are a

usable representation for biometric data. It targets iris codes, not face embeddings, and it protects a single database rather than routing a query among many databases.

3.2.9 Privacy-preserving face-based crowd monitoring

Marzani et al. (2026) in “Head count: privacy-preserving face-based crowd monitoring” [11] address counting individuals across locations without revealing their faces. They combine face features, fuzzy extractors and homomorphically encrypted Bloom filters so that the same person seen in different places is counted once while the underlying biometric stays hidden. This is the closest related work to using Bloom filters for cross-location face matching. Its purpose is counting, not identification and tracking of named occupants, and it has no role-based query layer; this project uses LSH codes and per-building Bloom filters to route a capture towards the building that holds the reference.

3.2.10 Space/time trade-offs in hash coding

Bloom (1970) in “Space/time trade-offs in hash coding with allowable errors” [12] introduces the Bloom filter: a bit array and several hash functions that answer “probably in the set” or “definitely not in the set”, with a controllable false-positive rate and no false negatives. The analysis trades a small error probability for a large saving in space. This is the property the project depends on: each building can publish a compact, one-way summary of its occupants. The filter answers exact membership only, which is why locality-sensitive hashing is needed to make nearby embeddings collide.

3.2.11 Summary Cache: Bloom filters for wide-area cache sharing

Fan et al. (2000) in “Summary Cache: a scalable wide-area web cache sharing protocol” [13] study how web proxies can share cache contents without flooding each other with queries. Each proxy keeps a Bloom-filter summary of the objects held by its peers and contacts only the peers whose summaries match, which sharply reduces inter-proxy messages at a small cost in false hits. The evaluation is on trace-driven simulations of proxy traffic. The same idea of summaries that tell where an item probably lives is applied here to faces, but a face capture never matches exactly, so the summaries must be built from similarity-preserving codes.

3.2.12 Similarity-preserving hashing: SimHash and multi-probe LSH

Charikar (2002) in “Similarity estimation techniques from rounding algorithms” [14] shows that random-hyperplane hashing maps vectors to bits so that the probability of two vectors agreeing on a bit grows with the angle between them, which makes binary codes

preserve angular similarity. Lv et al. (2007) in “Multi-probe LSH: efficient indexing for high-dimensional similarity search” [15] improve recall without more hash tables by also probing the buckets that differ from the query in the least confident bits. Kirsch and Mitzenmacher (2006) show that Bloom-filter positions can be derived by double hashing with no loss in asymptotic false-positive rate [16]. These three works supply the encoding, multi-probe lookup and hashing used in this project. They are generic indexing techniques and say nothing about biometric privacy or access control.

3.3 Comparison of reviewed literature

Table 2 compares the reviewed works on the attributes used above; the gap that follows from the comparison is stated below it.

Gap. None of the surveyed works routes a face from a building that does not know it to the building that does, using compact similarity-preserving summaries, and none combines this with role-based access to the tracking data and with explanations of the answers. That combination is the contribution of this project.

Table 2: Comparison of reviewed literature

Work	Problem or objective	Approach and contribution	Study context	Evaluation criteria	Key findings	Limitations and gap
[1] Hegde et al., 2024	Find when and where groups form indoors	Pose + clustering on edge cameras	One therapeutic space, 7 people, 35 min	Precision, recall, F1, MAE	F1 up to ~90%, MAE 1.25 m	Groups, not individuals; one site
[2] Kwon et al., 2023	Localize and track people with few cameras	39 edge units with pose estimation	~1,700 m² therapeutic space	Localization error, MOTA, MOTP, ID switches	~1.4 m error, MOTA ~88.6%	Anonymous tracking in one site
[3] Yang et al., 2022	Distributed multi-target association	Shared label and appearance table, no calibration	Benchmarks + own mobile-camera set	MOTA, IDF1, FP/FN, ID switches	Real-time association without a server	No building-level state or visitor origin
[4] Casao et al., 2021	Distributed multi-target tracking	Consensus Kalman filter + re-ID + tracker manager	Public multi-camera datasets	MOTA/IDF1-type, bandwidth	Better than centralised in accuracy and bandwidth	One camera network; no occupants or visitors
[5] Gaikwad and Karmakar, 2021	Edge multi-person multi-camera tracking	Detection + tracking + re-ID on Jetson TX2	Own campus recordings, one day	IDF1, FPS, re-ID latency	IDF1 90.97, ~30 FPS, ~90 ms	No multi-building state / query model
[6] Mohan and Menon, 2020	Model camera-network identification and tracking	Abstract DSTS: states, events, transitions	Abstract framework	Analytical	Formal model with building-level BSTS	Visitors via a central master list
[8] Nair et al., 2021	Face recognition and tracking	Recognition + tracking across cameras	Surveillance cameras	Recognition and tracking quality	Face is a workable identity cue	Single common reference store; no access control
[10] Rathgeb et al., 2014	Protect iris templates	Bloom-filter templates	Iris databases	Comparison and speed	Alignment-free, one-way templates	Iris, not face; one database, no routing
[11] Marzani et al., 2026	Privacy-preserving crowd counting	Face features + fuzzy extractor + encrypted Bloom filters	Multi-location counting	Counting accuracy, privacy	Counts people without revealing faces	Counts, does not identify; no roles
[12], [16] Bloom 1970; Kirsch and Mitzenmacher 2006	Compact set membership	Bit array + hash functions; double hashing	Theory	False-positive rate, space	No false negatives; tunable error	Exact membership only
[13] Fan et al., 2000	Share web caches cheaply	Bloom-filter summaries of peers	Proxy traces	Messages, hit ratio	Far fewer inter-proxy messages	Exact keys, not similar faces

Work	Problem or objective	Approach and contribution	Study context	Evaluation criteria	Key findings	Limitations and gap
[14], [15] Charikar 2002; Lv et al. 2007	Similarity-preserving hashing	Random hyperplanes; multi-probe lookup	Theory, similarity search	Recall, tables needed	Angle-preserving codes; fewer tables	Generic; no privacy or access control

Chapter 4
Requirements

4.1 Functional requirements

Table 3: Functional requirements

ID	Requirement	Actor	Acceptance criterion
FR1	Recognise a registered occupant against the building's own gallery	Camera event	Accepted when at least 12 of 20 references agree within 80.43°
FR2	Route an unrecognised capture to likely buildings using Bloom filters, then verify	Building node	A shortlist of at most 5 buildings; only they receive the capture
FR3	Update the person's zone probabilities after each identified event	System	Nine rows per event summing to 1
FR4	Remember visitors for a day and promote frequent visitors	System	Repeat sightings need no routing; promotion after 3 entries in 7 days or 8 in 30
FR5	Open and close a visit pointer at the home building	System	Pointer minted at arrival, closed at departure; no biometrics stored
FR6	Answer 13 query types and a full trajectory query	Dean, Teacher, Student	Answer consistent with the stored events
FR7	Apply a two-layer policy before any answer	Query service	Every denied request returns a reason code and no data
FR8	Return an explanation with each answer, redacted to the caller's level	Query service	Explanation never shows more than the answer
FR9	Seal every cross-building message	Building node	Tampered, replayed or mis-addressed messages are dropped
FR10	Show the campus, zones, queries and security log on a dashboard	All roles	Content filtered by the caller's role
FR11	Compare with a centralised search and report TP / FP / FN / TN	Evaluator	Metrics per cohort, per state level and per seed

4.2 Non-functional requirements

Table 4: Non-functional requirements

Quality	Requirement	Measure
Accuracy	Decentralised identification close to a centralised search	Overall accuracy within 1 point; visitor F1 reported
Communication cost	Touch a small part of the galleries per lookup	Gallery touched per lookup (central = 100%)
Privacy	No gallery, id or embedding in a published filter; no plaintext biometric between buildings	Filter contents inspected; sealed-transfer tests
Security	Fail closed; resist tampering, replay, impersonation, brute force	Attack tests; lockout after 5 failures in 60 s
Explainability	Explanations faithful to the answer and free of leakage	Per query class: a yes, a no and an empty case
Reproducibility	Every result regenerable from a seed	Seeded populations; five seeds with 95% CI
Performance	Run on a laptop CPU	11,000 events in about 8 minutes
Usability	One web page for all roles	Dashboard with role-filtered views

4.3 Hardware requirements

A laptop-class machine is sufficient: Intel CPU, 16 GB of memory recommended, and no GPU. All face-model inference is done once, offline, on the CPU; the experiments then work on stored 512-dimensional embeddings. No cameras or dedicated edge devices are required because camera events are replayed from the datasets.

4.4 Software and data requirements

Table 5: Software, tools and data

Item	Details
Operating system / language	Windows 11; Python 3.12 in a virtual environment
Numerical and data	NumPy, pandas, matplotlib; ONNX Runtime and OpenCV for the one-off embedding extraction
Face models	InsightFace buffalo_l: SCRFD detector (det_10g) [18] and ArcFace recogniser (w600k_r50) [17], 512-d output
Routing and state	Random-hyperplane LSH [14], Bloom filter [12], vote-based verifier, SQLite state stores (written from scratch)
Security and API	cryptography (X25519, HKDF, AES-GCM, Ed25519), HMAC tokens, FastAPI and uvicorn, Prometheus client
Front end	Static HTML / JavaScript dashboard served by the API (no build step)
Testing	pytest with 571 automated tests
Main dataset	CASIA-WebFace [19]: 500 identities, 20 reference + 20 test photos each (20,000 embeddings), 10 buildings of 50 people
Masked dataset	RMFRD [20]: 50 identities as an 11th building; 20 unmasked references + 20 masked probes each
Access conditions	Both datasets are public research datasets; RMFRD's page states no licence beyond a request to cite it

Dataset partitioning. For each person, the 20 reference photos enrol them in their building and are never used as events; the 20 test photos are the camera events, so no photo is used for both. Buildings are formed by a seeded random permutation of the 500 people into 10 buildings of 50; the 50 masked people always form the 11th building and are not shuffled by the seed.

Chapter 5
Proposed Methodology

5.1 Design (architecture)

The environment is a set of buildings. Each building node owns its registered occupants' gallery, a state store with a registered and a visitor database, and a visitor lifecycle manager. A secure communication layer connects the nodes; an access control layer sits between the users and the data; and a query service applies the policy and builds the response with its explanation. Figure 1 shows the architecture and Figure 2 the two main workflows.

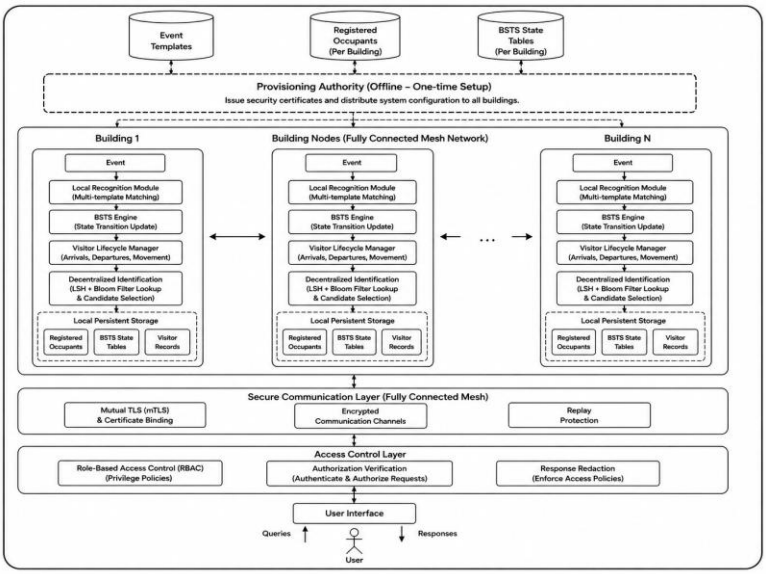


Figure 1: System architecture: building nodes, secure communication layer, access control layer and user interface

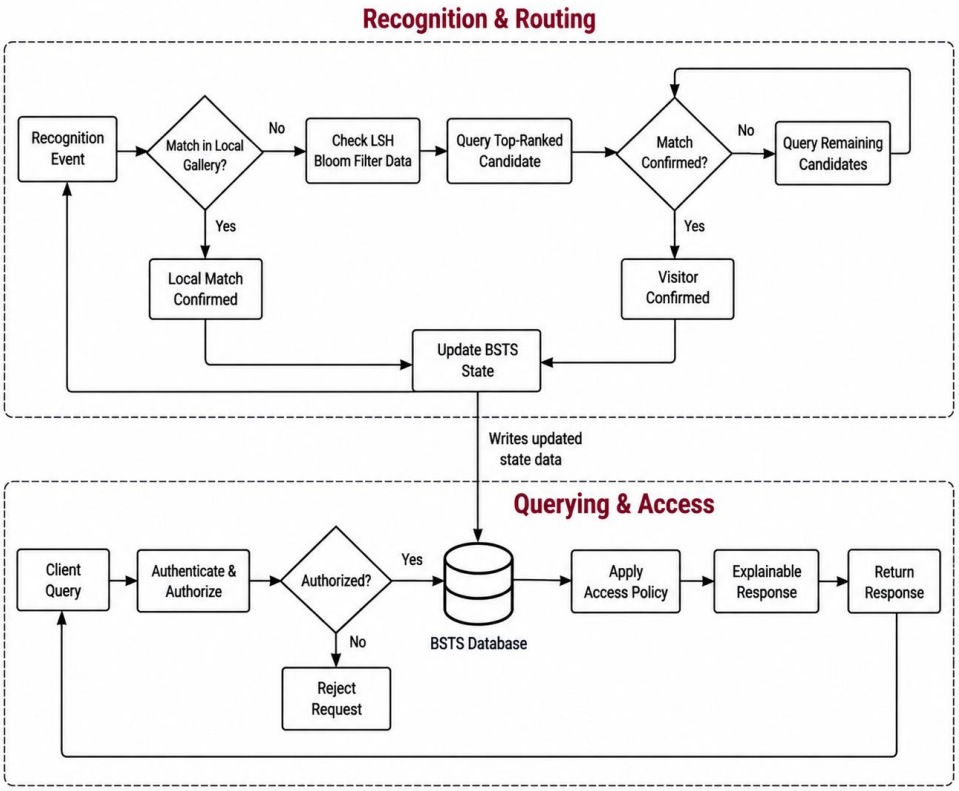


Figure 2: Inter-building tracking workflow and querying / access workflow

Novelty. On the technical side, (a) each building publishes only a Bloom filter of LSH codes, so a capture can rank buildings without any gallery leaving them, and (b) only the shortlisted buildings verify the face. On the application side, (c) a two-layer, class-teacher access policy with graded disclosure, (d) an explanation that is built from the same events as the answer and redacted to the caller's level, and (e) sealed inter-building transfers with certificate-bound keys.

5.2 Proposed approach and procedure

5.2.1 Embedding preprocessing and enrolment

Faces are detected and aligned with SCRFD [18] and embedded with an ArcFace model [17] (512 dimensions). A raw embedding is normalised, centred with the mean of the reference embeddings, and normalised again:

$$\hat{e} = \frac{e/\|e\| - \bar{x}}{\| e/\|e\| - \bar{x} \|} \tag{1}$$

The R = 20 preprocessed references of an occupant u are averaged into a unit-norm centroid:

$$z_u = \frac{\sum_{r=1}^{R} \hat{e}_{u,r}}{\left\| \sum_{r=1}^{R} \hat{e}_{u,r} \right\|} \tag{2}$$

5.2.2 LSH codes and the Bloom filter

Random hyperplanes (one Gaussian vector w for each slot l and bit k, drawn once and shared by all buildings) define L = 122 slots of K = 11 bits [14]. The bit h is the side of the hyperplane on which the embedding lies, and the K bits of slot l form an integer code c_l:

$$h_{l,k}(\hat{e}) = \mathbf{1}\left[w_{l,k}^{\mathsf{T}} \hat{e} \geq 0\right], \quad c_l = \sum_{k=0}^{K-1} h_{l,k}\, 2^k \tag{3}$$

Each occupant's centroid yields L items, the item for slot l being l times 2 to the power K, plus c_l, so a code counts as a match only in the same slot. The items of all occupants of a building are inserted into a Bloom filter [12] sized for 50 occupants at a target false-positive rate of 0.01. Only the packed bits and a few scalars are published, about 10 KB per building: no ids, embeddings or centroids.

5.2.3 Local recognition, routing and verification

A capture is first compared with the capturing building's own gallery. For every candidate occupant the angle between the capture and each of the 20 references is computed, and a reference votes if its angle is at most 80.43°. The winner is the occupant with most votes (ties broken by the smallest single angle); it is accepted only with at least 12 votes.

If the local check does not accept, the capture is hashed in all L slots. As well as the exact code, the 3 least confident bits of each slot are flipped to give extra probes [15]. Each peer's filter is tested; a building's score is the number of distinct slots that matched, on a 0 to L scale, and the top 5 buildings form the shortlist. They are contacted in rank order over the sealed channel and verify the capture with the same vote rule against their own occupants; the first confirmation ends the search. If no one confirms, the capture is not identified (abstain).

5.2.4 Tracking: probabilities and state

For an identified event with candidate distances d_i to each candidate occupant i, the occupant probabilities follow Definition 3.3 of the DSTS framework [6]: with σ the population standard deviation of the distances,

$$p_i = \frac{\exp\left(-d_i^2 / 2\sigma^2\right)}{\sum_j \exp\left(-d_j^2 / 2\sigma^2\right)} \tag{4}$$

(uniform over ties when σ = 0). The Bloom score, vote counts and ground truth never enter this probability. The building state store keeps, for each occupant, a probability over the

nine zones (z1 Entrance, z2 Mail room, z3 Office, z4 Lounge, z5 Conference, z6 Classroom, z7 Cafeteria, z8 Exit, and zT, the transition zone to the outdoors) and updates it from each event; movement between buildings must pass through zT. The event is stored in the building that observed it: registered occupants in the registered database and visitors in the visitor database.

5.2.5 Dynamic visitor database and visit pointers

A visitor is remembered in three layers. *Presence* lists who is inside and in which zone, until they leave or the day ends. The *day cache* keeps the visitor's 20 references for one simulated day, so that later sightings the same day are matched locally without routing. The *entry history and permanent pool* record every arrival; when earlier entries already satisfy a rule (default: 3 entries in 7 days, or 8 in 30) the next entry stores the visitor's embeddings permanently. A *transition-zone rule* keeps a visitor who is waiting in zT out of other people's probability updates until they are seen outside zT. A *visit pointer* (occupant id, current building, entry and exit times, no biometrics) is minted at the home building when the visitor arrives and closed at departure, so a location query follows the pointer instead of asking all buildings.

5.2.6 Role-based access control

Every query passes two layers before execution. *Layer 1* authenticates the principal and applies the building scope. *Layer 2* is the persona policy, which returns a disclosure level for the pair (caller, target) and the purpose and time of the query. The levels are L0 none, L1 presence, L2 current sector with the time rounded to the hour, L3 exact current zone, and L4 full historical track. Table 6 states the access model.

Table 6: Access model (class-teacher RBAC)

Role	May follow	Disclosure	Cannot see
Dean (1 per building)	Teachers and students of their own campus (any building they are in), and other Deans	L4 for own campus; coarse for other Deans	Other campuses' teachers and students; visitors
Teacher (10 per building)	Only the students on their class roster, in any building	L4 for roster students	Unassigned students, peer teachers, the Dean
Student (39 per building)	Themselves in full; the teachers assigned to them	L4 self, L3 assigned teachers	Peer students, other teachers, the Dean

Building-level queries (Q1, Q2) return aggregates to everyone in their own building and the list of people only to a Dean; zone queries (Q10, Q12, Q13) filter each listed occupant through the same policy; students cannot run zone queries. The thirteen templates are: Q1 did anyone stay after T; Q2 were there more visitors than registered occupants at T; Q3 did O leave before T; Q4 did O enter between t1 and t2; Q5 did O visit all zones; Q6 where

was O at T; Q7 how many zones did O visit in an interval; Q8 when did O first visit zone z; Q9 how long did O spend there; Q10 who was in zone z at T; Q11 where was O during an interval; Q12 who was in zone z during an interval; Q13 who ever visited zone z; plus TRACK, the full trajectory.

5.2.7 Explainable responses

Each query builds an explanation from the same filtered event list that it used to compute the answer, never by recomputing afterwards. The explanation holds the decision rule in one sentence, the evidence events with their roles (decisive, last before t, carry-in state, zone visit), what was looked for and not found, uncertainty flags, what redaction withheld, and the visit pointer if one answered the query. Flags come from the identification provenance stored with every identified event (votes, routing rank, Bloom margin, whether the vote winner was also the nearest candidate): low votes (fewer than 16), vote winner not nearest, verified by a lower-ranked building, narrow routing margin (below 5), stale observation (older than 60 minutes) and pointer-based. The identity probability is not used as a confidence signal because it is close to 1 for most misidentifications too. Before leaving the service the explanation is redacted to the caller's disclosure level.

5.2.8 Secure inter-building exchange

Every message between buildings is sealed. The sender generates an ephemeral X25519 key [23], derives a symmetric key with HKDF-SHA256 [22], encrypts with AES-GCM [25] using the whole header as associated data, and signs the header, nonce and ciphertext with Ed25519 [24]. Replies are bound to their request. A campus certificate authority issues node certificates that bind both the signing key and the key-agreement key; expiry, revocation, subject and pinning are checked. A replay guard accepts a message only within a 300-second window and once per nonce, and runs after the signature check so the timestamp is authenticated. Any failed step yields no reply and nothing is admitted (fail closed). The API issues HMAC-signed tokens that expire after one hour, derives a separate login credential for each occupant from a master secret, locks out a client after 5 failures in 60 seconds, and replaces the WebSocket token in the URL with a single-use ticket.

5.2.9 The masked building

To study masked faces, 50 more people from RMFRD [20] form an 11th building. The 50 people with the most real masked photos are used. Each has 20 distinct real unmasked photos as the gallery and 20 masked probes: all their real masked photos up to 20 (800 of the 1,000 probes), topped up with augmented copies (flip, rotation, scale, shift, brightness,

blur, noise, JPEG compression) for the 200 remaining. Photos are detected and aligned with SCRFD and embedded with the same ArcFace model. Routing and verification are unchanged: a masked probe is matched against an unmasked gallery. The masked people are not shuffled by the seed.

5.2.10 Parameters

Table 7: Parameters used

Parameter	Value
LSH bits per slot K / slots L	11 / 122
Bloom target false-positive rate	0.01
References per occupant R	20
Accept angle / minimum votes	80.43° / 12
Shortlist size / extra probes	5 / 3
Embedding dimension	512
Visitor promotion rule	3 entries in 7 days or 8 entries in 30 days
Replay window / token lifetime	300 s / 1 h
Login lockout	5 failures in 60 s per client and occupant, or 20 in 5 min per occupant
Contract seed	42

Chapter 6
Results and Analysis

6.1 Evaluation setup

The evaluation uses 550 occupants in 11 buildings: the 500 CASIA-WebFace people dealt into 10 buildings of 50 by a seeded permutation, and the 50 masked people in building 11. One simulated day has 11,000 events, one for each of the 11,000 test photos; 8,800 are local (the person is in their home building) and 2,200 are visitor events. The main run uses seed 42; variation is measured on five further seeds (0 to 4), each of which places the same people in different buildings and draws a different event day. The baseline is a hypothetical centralised server that holds every building's gallery and searches all of them with the same preprocessing and vote rule, without LSH or Bloom filters. Ground truth is recorded for each event and, for every state, as a full snapshot of who is in which building and zone, so that tracking can be scored at every step.

6.2 Evaluation criteria

All metrics use the one-vs-rest convention. An item has one true label and at most one predicted label out of L labels. **TP**: the predicted label is the true label. **FP**: a wrong label was predicted. **FN**: the true label was not predicted (a wrong prediction or an abstention). **TN**: every other (item, label) pair, that is items × L − TP − FP − FN. A wrong identity is therefore both an FP and an FN, and an abstention only an FN.

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R} \tag{5}$$

Identification accuracy is the share of events whose predicted identity is the true one. Because TN is large, accuracy computed with TN is close to 1 and is not used to compare systems. **Event level**: the label is the occupant id (500 + 50 labels), reported for all, local, visitor, masked and unmasked events. **State level**: state s_k is the campus after event k; the label is (building, zone) with 99 labels, plus building (11 labels) and zone (9 labels) alone, and every occupant the system places counts, including visitors updated by other people's events. **Cost**: gallery touched per lookup is the number of buildings whose gallery is searched divided by the number of buildings (the centralised server is 100%); routing recall is the share of routed visitors whose home building is in the shortlist.

6.3 Result

6.3.1 Identification: decentralised against centralised

Table 8: Identification accuracy and gallery touched, seed 42, 11,000 events

Cohort	Events	Centralised accuracy	Decentralised accuracy	Gallery touched
All events	11,000	92.24%	91.76%	10.3%
Local	8,800	92.03%	92.31%	9.1%
Visitor	2,200	93.05%	89.59%	15.2%
Unmasked people	10,000	95.48%	95.04%	10.1%
Masked people	1,000	59.80%	59.00%	12.3%

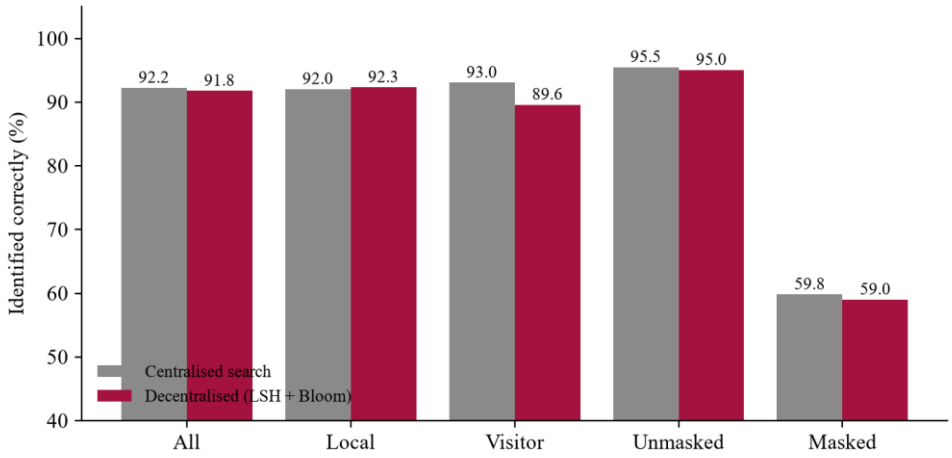


Figure 3: Identification accuracy, centralised search against the decentralised pipeline (seed 42)

Over all events the decentralised pipeline identifies 91.76% against 92.24% for the central search, a difference of 0.47 points, while each lookup touches 10.3% of the galleries. Local events are slightly better with the pipeline (92.31% against 92.03%) because a person is matched only among the 50 people of their own building, so there are fewer look-alikes to confuse them with; the central search must discriminate among 550. The visitor events lose 3.5 points (89.59% against 93.05%), the cost of routing.

Table 9: Event-level confusion counts and scores, seed 42

Cohort	TP	FP	FN	TN	Precision	Recall	F1
All	10,094	217	906	6,038,783	0.9790	0.9176	0.9473
Local	8,123	134	677	4,831,066	0.9838	0.9231	0.9525
Visitor	1,971	83	229	1,207,717	0.9596	0.8959	0.9267
Unmasked	9,504	147	496	5,489,853	0.9848	0.9504	0.9673
Masked	590	70	410	548,930	0.8939	0.5900	0.7108

6.3.2 Routing and the visitor database

Of the 2,200 visitor events, 1,447 (65.8%) are served from the building's visitor database (the same visitor sighted again the same day) and 708 are routed. Routing recall, the share

of routed visitors whose home building is in the top five, is 85.3%, with 2.08 buildings contacted per routed attempt. Visit-pointer precision is 98.6%. Figure 4 breaks the visitor events down by outcome: errors are mostly routing misses that end in an abstention, and a smaller number of cases where the right building is reached but the face still does not pass verification.

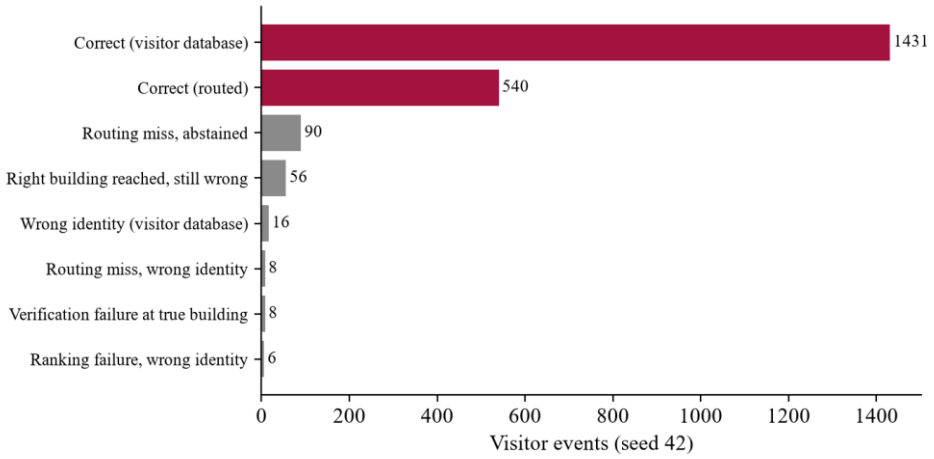


Figure 4: Outcome of the 2,200 visitor events (seed 42)

6.3.3 Stability across five random populations

Table 10: Results over five random populations (one simulated day, 11,000 events each)

Metric (mean over seeds 0-4)	Mean	Std	95% CI (±)	Min	Max
Identification accuracy, all events	91.54%	0.16	0.20	91.35%	91.76%
Identification accuracy, local	92.02%	0.17	0.21	91.75%	92.18%
Identification accuracy, visitors	89.61%	0.56	0.69	88.86%	90.36%
Routing recall (top 5)	86.86%	0.32	0.40	86.42%	87.14%
Event F1, unmasked people	0.9646	0.0018	0.0023	0.9624	0.9671
Event F1, masked people	0.7106	0.0026	0.0032	0.7069	0.7131
State F1, building	0.9501	0.0040	0.0049	0.9437	0.9533
State F1, zone	0.9155	0.0021	0.0026	0.9128	0.9180
State F1, building + zone	0.8957	0.0030	0.0037	0.8904	0.8976

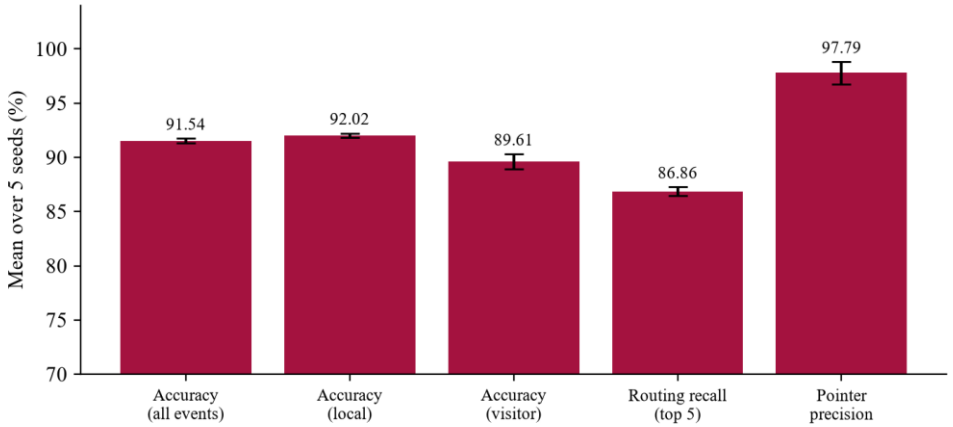


Figure 5: Mean over five seeds with 95% confidence intervals

The spread is small: overall accuracy is 91.54% with a 95% confidence interval of ±0.20 points, so the seed-42 result is representative and does not depend on one lucky assignment of people to buildings. The visitor results vary more (±0.69 points) because they rest on fewer events.

6.3.4 Tracking state

Table 11: State-level scores, mean over all 11,000 states (seed 42)

Level	Labels	Precision	Recall	F1	Visitor recall
Building + zone	99	0.902	0.892	0.897	0.908
Building	11	0.958	0.947	0.952	0.937
Zone	9	0.919	0.908	0.913	0.946

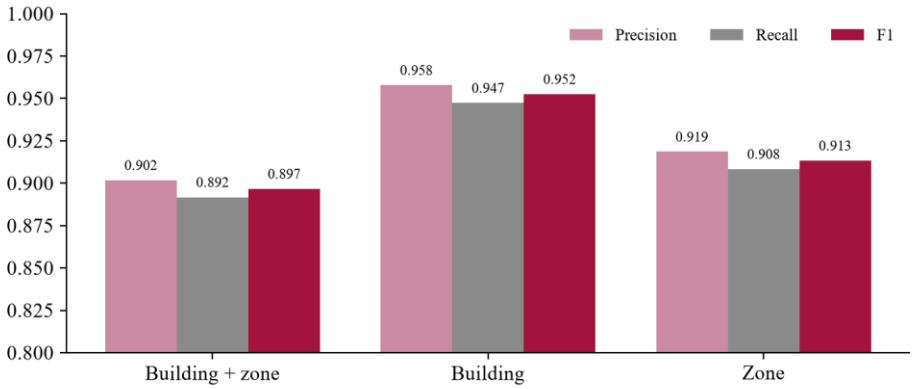


Figure 6: Precision, recall and F1 of the tracked state at three levels

Building-level state, that is knowing which building each person is in after every event, reaches an F1 of 0.952; the zone within a building and the combination are harder because the nine-zone belief is updated from one event at a time.

6.3.5 Masked people

Table 12: Masked against unmasked people (seed 42)

Group	Events	Identified correctly	Abstained	Precision	F1
Unmasked people	10,000	95.04%	349	0.985	0.967
Masked people	1,000	59.00%	340	0.894	0.711
Masked, at home	850	59.41%	-	0.895	0.714
Masked, visiting	150	56.67%	-	0.885	0.691

Masked people are identified 59.0% of the time against 95.0% for unmasked people; 340 of the 1,000 masked captures are left unidentified. When the system does answer for a masked person it is right 89.4% of the time, so the loss is mostly missed identifications rather than wrong ones. A masked probe is about 72° from its own unmasked gallery on average, close to its nearest other person, so the gallery gives little margin. Routing is also weaker for masked visitors: the right building is found less often and ranked lower.

What was tried. Three ideas were tested offline on the 1,000 masked probes against the 50-person gallery, using the real thresholds. (i) Adding ten synthetic-masked templates per person to the gallery; (ii) matching on the upper face only, by blanking the lower part of both gallery and probe; (iii) relaxing the accept angle and votes when a mask is suspected. Each mostly traded abstentions for wrong answers: relaxing the thresholds alone raised wrong identities from 7.0% to 28.4% and let strangers be accepted 31% of the time. F1 stayed at 0.72 or fell (Figure 7). Without any threshold, the nearest person is the right one only 65-70% of the time whatever the gallery, so the limit is the face model, which was not trained for masks, and the noisy web labels. Enrolling five to eight real masked photos per person gave about five points more in rank-1 accuracy.

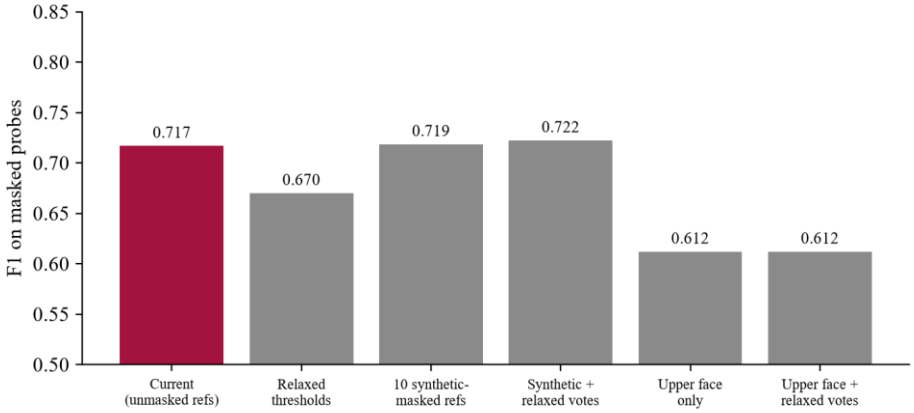


Figure 7: F1 of masked probes for the tested variants (offline, building 11)

The existing mask handler. The pipeline already contained a handler that flags a probe as masked when the spread of per-person minimum angles is large, and then loosens the angle by 4° and the votes by 25%. Measured on seed 42 it flags none of the 850 masked local events and 3,355 of the 7,950 unmasked ones (42%), because a large spread means one person stands out, not that a mask is present. With the handler switched off (DSTS_MASK_HANDLER=off), masked results are unchanged and unmasked people gain 21 correct identifications and lose 22 false accepts. The default is left on and the finding is reported as a limitation.

6.3.6 Verification of the security and policy layers

Table 13: Automated tests by area

Area	Tests	What is checked
access_control	112	Every policy cell, structured Layer-1 decisions, building scope, lockout, credentials, tickets, API-level RBAC attacks
secure_transfer	61	Sealed payloads: tampering with every header field, replay, wrong sender or recipient, revoked certificate, fail closed, strict decoding

Area	Tests	What is checked
api	92	Login, queries per role, caching, zone and time-frame views, dashboard adapter
queries	184	Query engine, multi-day normalisation, explanation faithfulness, leakage and uncertainty, CLI access rules
pipeline, population, events, evaluation	98	Phase 2/3/4, visitor database, seeded populations, event generation, ground truth, confusion metrics
standalone, centralised, dsts	19	Standalone demo, centralised baseline, state library
shared utilities	5	Dated timestamps, time-window arithmetic
Total	**571**	All passing on the final commit

A persona matrix tool replays 64 requests (Dean, Teacher, Student and an outsider against each query type and target type) and checks the decision, rule code and disclosure level of each. Examples: a Teacher asking for an unassigned student or for the Dean gets RULE_B_UNASSIGNED_DENIED (L0); a student asking for a peer gets RULE_C_PEER_STALKING_DENIED; a student asking for a zone listing gets RULE_C_PEER_ZONE_SURVEILLANCE_DENIED; any cross-building request gets a CROSS_BUILDING_DENIED rule; a person asking for themselves gets RULE_SELF_LOOKUP at L4.

6.3.7 The dashboard



Figure 8: Campus overview: 11 buildings and 550 occupants, with building 11 marked as the all-masked building

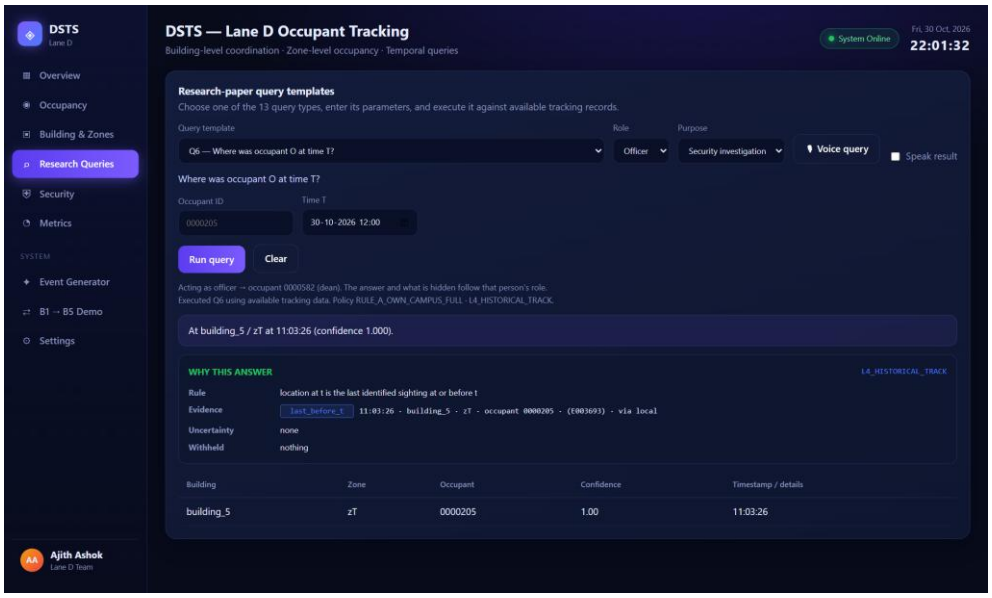


Figure 9: A Dean's query with its "Why this answer" card (rule, evidence, uncertainty, withheld)

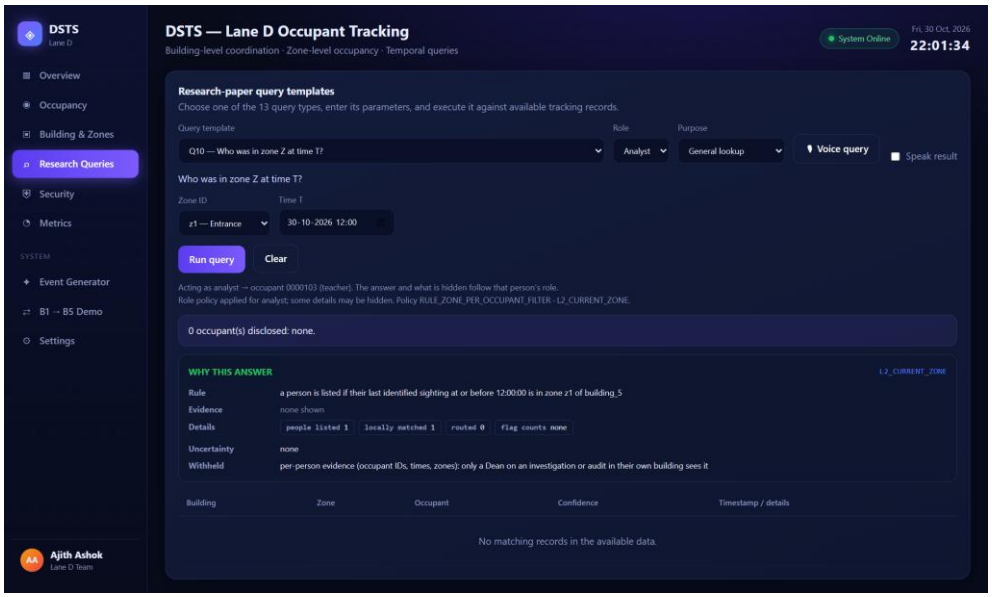


Figure 10: The same page for a Teacher: the answer is reduced and the card states what was withheld

6.4 Comparative analysis

Table 14: Comparison with the baseline and the approaches in the literature

Approach	Visitor lookup	Face data leaving a building	Result here
Centralised search over all galleries (baseline)	One global search	Galleries pooled in one store	92.24%; 100% of the gallery searched
Decentralised LSH + Bloom routing (this work)	Rank by Bloom score, verify top 5	Only a 10 KB Bloom filter; references move only after a confirmed match	91.76%; 10.3% of the gallery
DSTS base paper [6]	Central master list	Master list of visitor embeddings	Not re-run; replaced by the above
Edge-camera tracking [1]-[5]	Within one site	n/a	No cross-building identity, RBAC or explanations

Other published systems [1]-[5] use different data and tasks and cannot be compared number for number; the fair comparison is the same pipeline with a central search. Routing costs 0.47 points overall and 3.5 points on visitors, in exchange for touching about a tenth of the gallery and never pooling it.

6.5 Discussion and interpretation

The results meet the objectives that can be measured. Decentralised identification is within half a point of a central search overall while touching about 10% of the galleries, so the privacy and locality gain costs little accuracy (objectives 1 and 4). Most visitor events are cheap because repeat sightings the same day come from the day cache, which is why the number of buildings contacted per visitor event is small. Tracking is reliable at building level and good at zone level (objective 1). The access model and the explanations are verified by tests rather than by accuracy figures (objectives 2, 3 and 5): every policy cell and every query class is exercised, and an explanation is checked to be faithful to its answer and to reveal no more than the answer does. The weakest result is masked identification, where the limiting factor is the face model rather than the system design.

6.6 Limitations

- The network, cameras and the face detector are simulated. Arrays are plaintext in memory before sealing and after opening, and the certificate authority is a stand-in whose signature is a message authentication code, not a real public-key infrastructure.
- The API has no TLS unless a reverse proxy adds it, no browser security headers, and no encryption of the stored embeddings and state databases. No outside penetration test or fuzzing was done; the attack tests are those the team wrote.
- Token secrets, caches and lockout counters are per process, so they would not hold across several servers; the dashboard's demo mode lets any viewer act as any role by design.
- The masked building uses web-crawled photos with some wrong labels, and 20% of the masked probes are augmented copies. Masked results are not an accuracy estimate for new people.
- A wrong identity stored in the visitor cache stays for the rest of that day, and a visitor whose departure is not identified stays “present” until the day ends.

- Only 550 people in 11 buildings were measured; larger scale is untested. The same 20 test photos per person are reused across days in multi-day runs.
- Results are from simulated movement on replayed photos, not from live video.

Chapter 7
Conclusion

7.1 Conclusion

This project built and evaluated a system that identifies and tracks occupants across buildings without pooling face data, answers queries according to the caller's role and relationship, and explains each answer. A visitor is found by routing a capture with LSH codes and per-building Bloom filters and verifying it only in the shortlisted buildings. On 11,000 events it identifies 91.8% correctly against 92.2% for a centralised search, touching 10.3% of the galleries; over five random populations the accuracy is 91.5% ± 0.2. A dynamic visitor database makes most repeat visits free of routing. A two-layer class-teacher policy, sealed transfers and redacted explanations are implemented and covered by 571 automated tests. Masked people are the hard case at 59.0%, and the evidence points to the face model, not the routing, as the limit. All six objectives are achieved in simulation; the security work is a tested design on a simulated network and not a hardened deployment.

Table 15: Objective-wise status

Objective	Planned	Status	Evidence
1 Multi-building DSTS extension	Mid review	Achieved	BSTS over 11 buildings; building-level state F1 0.9501
2 RBAC query interface	24/9/26	Achieved	Two layers, class-teacher model, every policy cell tested
3 Explainable responses	26/9/26	Achieved	Rule, evidence, flags, redaction; faithfulness tests
4 LSH + Bloom routing	Mid review	Achieved	10.3% of gallery touched, 0.47-point accuracy cost
5 Security layer	2/10/26	Achieved (simulated network)	Sealed transfers, tokens, lockout, audit log
6 Evaluation	Final review	Achieved	Event and state TP/FP/FN/TN, five seeds, centralised baseline

7.2 Future work

- **Masked recognition.** Use a face model trained with masks, replace the angle-spread mask heuristic with a real mask detector, and enrol a few real masked photos per person (about five points of rank-1 accuracy in our tests).
- **Tracking context.** Prefer people known to be inside the building and fuse consecutive frames, to reduce masked errors; validate on the same seed-42 events.

- **Production security.** Real TLS between buildings, a real public-key infrastructure, encryption at rest, security headers, and an independent penetration test and fuzzing of the decoder and API.
- **Real deployment.** Live cameras and detectors, more buildings and people, and a multi-week run to test the permanent visitor pool.
- **Learned routing.** Compare the LSH + Bloom routing with learned or partitioned routing for distributed vector search.

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